EuroMCM 2026 Problem λ : The Switching Point



La Gare Saint-Lazare by Claude Monet, 1877

1 Track Notes

Problem λ is the Modern AI problem, though it also touches on the content of the Operations Research/ Network Science (δ), Sustainability (Ω) and Policy (θ) tracks. The track label describes the problem, not the methods a team is expected to use: solutions that draw on methods associated with any EuroMCM track are equally acceptable, and the choice of approach has no bearing on how a paper is judged. Creative solutions are encouraged.

2 Background

The steam engine transformed industry by converting fuel into mechanical work. On the railways, this power had to be directed through a shared network. At a junction, a railway switch sends a train from one track to another. The route chosen affects journey time, congestion, and resource use, and the switch must be set before the train reaches the junction.

AI services face a similar decision. A routine request may be handled by a compact large language model (LLM), while a demanding task may require a more capable model or additional reasoning. At this computational switching point, the system must decide which path a request should take before it knows whether the answer will be satisfactory. Using the strongest model for every request can waste resources; relying only on the least expensive option can undermine answer quality.

These decisions also have environmental consequences. The electricity consumed by AI inference accumulates across millions of requests, while the carbon intensity of electricity varies between European regions and over time. Moving a computation to a lower-carbon location or waiting for cleaner electricity may reduce emissions, but can also increase response time. Moreover, a cheaper and more efficient AI service may attract enough additional demand to offset its expected environmental savings.

For this problem, imagine a hypothetical public-sector coordination body, the Sustainable Workloads and Inference Technology Coordination Hub (SWITCH). It brings together public agencies exploring how to use AI in public services. SWITCH is considering a shared framework for allocating inference

resources. Requests may be directed to different models, given additional reasoning time, or processed at different computing locations. The aim is to provide enough intelligence without unnecessary computation. How should the system decide when a different path is worth taking?

3 Requirements

Within this scenario, your team advises SWITCH. Develop and assess an intelligent routing strategy that balances the usefulness of AI responses with the resources required to produce them. Your solution should address the following aspects:

- Establish a framework for comparing the quality, computational cost, response time, and environmental impact of AI responses across different types of requests. How can answer quality or task difficulty be estimated when a correct answer is not always available for direct comparison?
- Design and evaluate an adaptive AI router that decides whether a request should use a lightweight model, a more capable model, or additional inference-time computation. Investigate how the router could learn from previous outcomes, recognise uncertainty, or revise an unsatisfactory response. Compare its performance with simpler, non-adaptive alternatives.
- Explore how routing decisions should change when demand, server availability, and electricity carbon intensity vary. Consider whether some requests should be processed at a different time or location, and identify the trade-offs between answer quality, waiting time, and carbon emissions. Under what conditions does more sophisticated routing cease to be worthwhile?
- Test the robustness of your proposed system when requests become more difficult, predictions are unreliable, or usage grows unexpectedly. Could an apparently efficient router increase total emissions or systematically disadvantage certain types of requests? Identify the information and evidence needed to evaluate its claims of environmental benefit.

Teams may use accessible AI models, public benchmarks, published measurements, or reproducible simulations, provided that assumptions and limitations are clearly explained. The central focus should remain on the intelligent decision-making mechanism and its evaluation, rather than carbon accounting alone.

4 Share Your Insights

Prepare a one-page, non-technical recommendation memo for SWITCH's governing board. Explain whether the proposed routing strategy should be adopted, what makes its decisions better than a fixed model-selection policy, and what trade-offs or failure modes participating public agencies should monitor. Present your principal findings in a form suitable for both technical management and an informed public audience.

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5 Submission

Your PDF solution (≤25 pages) should include:

- One-page Summary Sheet
- Table of Contents
- Complete Solution
- One-page Recommendation Memo
- References
- Report on Use of AI Tools (if applicable; excluded from the 25-page limit)

There is no mandatory minimum length. Teams may submit incomplete solutions. The use of AI tools is allowed but optional; compliance with EuroMCM AI usage policy is required.

6 Links

- RouteLLM: Learning to Route LLMs with Preference Data
- FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance
- International Energy Agency: Energy and AI
- ENTSO-E Transparency Platform
- European Environment Agency: Greenhouse Gas Intensity of Electricity Generation

7 Glossary

Adaptive Inference: Adjusting the model or amount of computation used for a request according to its estimated needs.

Model Cascade: A strategy in which requests may pass through multiple AI models, with more capable models invoked when necessary.

Quality Estimation: Predicting how useful or reliable a model's answer is likely to be, possibly without access to a definitive correct answer.

Grid Carbon Intensity: The greenhouse gas emissions associated with generating a unit of electricity, commonly expressed in gCO_2e/kWh.

Rebound Effect: An increase in usage following an efficiency improvement that offsets some or all of the expected resource savings.

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